PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\Asus\Desktop\mlops-recommender> & c:\Users\Asus\Desktop\mlops-recommender\venv\Scripts\Activate.ps1
(venv) PS C:\Users\Asus\Desktop\mlops-recommender> python src/onnx_convert.py
Model trained successfully!
ONNX model saved at models/model.onnx
(venv) PS C:\Users\Asus\Desktop\mlops-recommender>
```

hybrid_model.py | main.py | monitor.py | __init__.py | retrain.py | index.html | onnx_convert.py | Dockerfile

Dockerfile > ...

```
# Use official Python image
FROM python:3.10-slim

# Set working directory
WORKDIR /app

# Copy all files
COPY . .

# Install dependencies
RUN pip install --no-cache-dir pandas numpy scikit-learn fastapi uvicorn skl2onnx

# Expose port
EXPOSE 8000

# Run the API
CMD ["uvicorn", "api.main:app", "--host", "0.0.0.0", "--port", "8000"]
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

History restored

```
PS C:\Users\Asus\Desktop\mlops-recommender> & c:\Users\Asus\Desktop\mlops-recommender\venv\Scripts\Activate.ps1
(venv) PS C:\Users\Asus\Desktop\mlops-recommender> docker run -p 8000:8000 recommender-app
failed to connect to the docker API at npipe:////./pipe/dockerDesktopLinuxEngine; check if the path is correct and if the daemon is
 running: open //./pipe/dockerDesktopLinuxEngine: The system cannot find the file specified.
(venv) PS C:\Users\Asus\Desktop\mlops-recommender> docker run -p 8000:8000 recommender-app
INFO:     Started server process [1]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
INFO:     Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
```

docker desktop PERSONAL

Search Ctrl+K

Gordon BETA
Containers
Images
Volumes
Kubernetes
Builds
Docker Hub
Docker Scout
Models
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Extensions

Containers Give feedback

Container CPU usage
0.19% / 1600% (16 CPUs available)

Container memory usage
224.4MB / 7.45GB

Show charts

Search

Only show running containers

Name	Container ID	Image	Port(s)	CPU (%)	Last started	Actions
condescending_ardinghel	b7ae7d8fe97e	recommender-app	8000:8000	0.15%	13 seconds ago	

Showing 1 item

Walkthroughs

Multi-container applications
8 mins

Containerize your application
3 mins

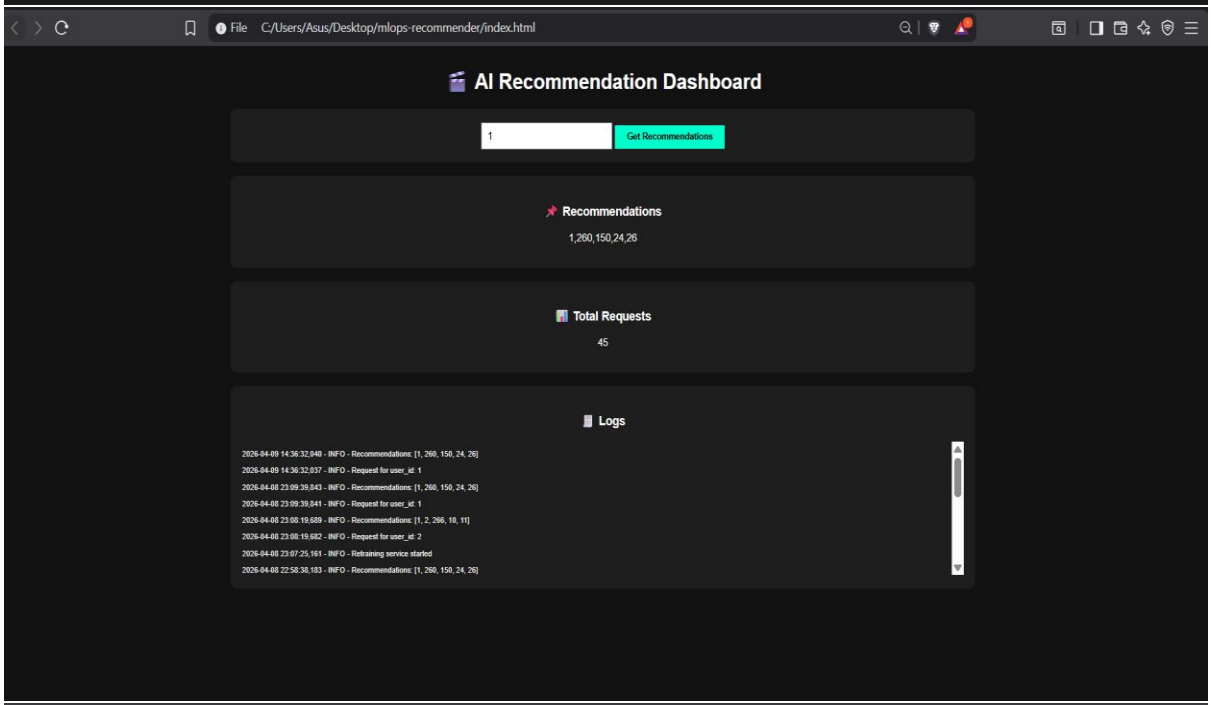
View more in the Learning center

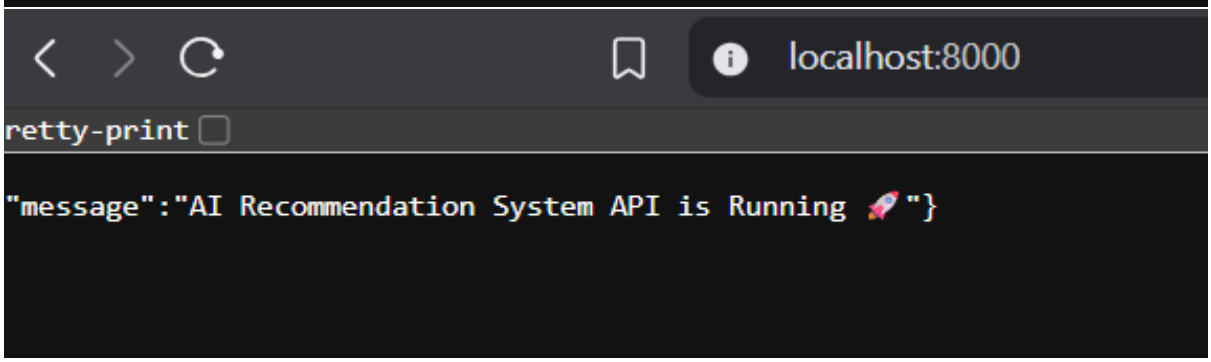
Upgrade plan

Engine running RAM 1.70 GB CPU 0.12% Disk: 6.91 GB used (limit 1006.85 GB) Terminal v4.68.0

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
(venv) PS C:\Users\Asus\Desktop\mlops-recommender> python -m uvicorn api.main:app --reload
INFO:     127.0.0.1:50114 - "GET / HTTP/1.1" 200 OK
INFO:     127.0.0.1:50114 - "GET /favicon.ico HTTP/1.1" 404 Not Found
2026-04-09 14:36:32,037 - INFO - Request for user_id: 1
2026-04-09 14:36:32,040 - INFO - Recommendations: [1, 260, 150, 24, 26]
INFO:     127.0.0.1:49167 - "GET /recommend?user_id=1 HTTP/1.1" 200 OK
INFO:     127.0.0.1:49167 - "GET /stats HTTP/1.1" 200 OK
INFO:     127.0.0.1:51102 - "GET /logs HTTP/1.1" 200 OK
INFO:     127.0.0.1:60869 - "GET /metrics HTTP/1.1" 404 Not Found
INFO:     127.0.0.1:60869 - "GET /favicon.ico HTTP/1.1" 404 Not Found
2026-04-09 14:39:40,615 - INFO - Request for user_id: 1
2026-04-09 14:39:40,618 - INFO - Recommendations: [1, 260, 150, 24, 26]
INFO:     127.0.0.1:56765 - "GET /recommend?user_id=1 HTTP/1.1" 200 OK
```





EXPLORER

MLOPS-RE...

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  - retrain.py
  - test_data.py
- venv
- Dockerfile
- index.html

OUTLINE

TIMELINE

test_data.py | data_transform.py | model.py | hybrid_model.py | main.py | monitor.py | __init__.py | retrain

api > main.py > recommend

```python
# ✅ FORCE add project root to Python path (FINAL FIX)
sys.path.append(os.path.abspath(os.path.join(os.path.dirname(__file__), "..")))

from fastapi import FastAPI
from fastapi.middleware.cors import CORSMiddleware

from src.hybrid_model import hybrid_recommend
from src.logger import get_logger

# Initialize FastAPI app
app = FastAPI()
app.include_router(monitor_router)

# Logger setup
logger = get_logger()

# Enable CORS
app.add_middleware(
    CORSMiddleware,
    allow_origins=["*"],
```

PROBLEMS | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS

```
(venv) PS C:\Users\Asus\Desktop\mlops-recommender> python src/hybrid_model.py
Hybrid Recommendations for User 1:
[1, 260, 150, 24, 26]
(venv) PS C:\Users\Asus\Desktop\mlops-re                    in:app --reload
INFO:     Will watch for changes in these directories: ['C:\Users\Asus\Desktop\mlops-recommender']
INFO:     Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO:     Started reloader process [15224] using StatReload

Hybrid Recommendations for User 1:
[1, 260, 150, 24, 26]
INFO:     Started server process [14784]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
```

Navigate to Command
Scroll to Previous Command (Ctrl+UpArrow)
Scroll to Next Command (Ctrl+DownArrow)

python | powers | powers

EXPLORER

MLOPS-RECOMMENDER

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    - logger.cpython-314.pyc
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OUTLINE

TIMELINE

test_data.py | data_transform.py | model.py | hybrid_model.py | main.py | monitor.py | __init__.py | retrain

src > hybrid_model.py > ...

```python
# HYBRID RECOMMENDATION FUNCTION
# ------------------------------

def hybrid_recommend(user_id, top_n=5):

    # Collaborative recommendations
    similar_users = user_similarity_df[user_id].sort_values(ascending=False)[1:6]

    collab_products = set()
    for sim_user in similar_users.index:
        products = user_item_matrix.loc[sim_user]
        top_products = products[products > 0].sort_values(ascending=False).head(5)
        collab_products.update(top_products.index)

    # Content-based recommendations
    user_products = data[data["user_id"] == user_id]["product_id"].values

    content_products = set()
    if len(user_products) > 0:
        last_product = user_products[-1]
        similar_items = get_similar_items(last_product)
        content_products.update(similar_items)
```

PROBLEMS | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS

```
(venv) PS C:\Users\Asus\Desktop\mlops-recommender> python src/hybrid_model.py

Hybrid Recommendations for User 1:
[1, 260, 150, 24, 26]
(venv) PS C:\Users\Asus\Desktop\mlops-recommender>
```

powershell | powershell | powershell

test_data.py | data_transform.py | model.py | hybrid_model.py | main.py | monitor.py | __init__.py | retrain

src > data_transform.py > ...

```python

# Load dataset
ratings = pd.read_csv("data/ratings.csv")

# Take subset for performance
ratings = ratings.head(100000)

# Convert ratings to interactions
def convert_interaction(rating):
    if rating >= 4:
        return "purchase"
    elif rating == 3:
        return "click"
    else:
        return "view"

ratings["interaction"] = ratings["rating"].apply(convert_interaction)

# Rename columns to match e-commerce format
ratings.rename(columns={
    "userId": "user_id",
```

PROBLEMS | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS

```
(venv) PS C:\Users\Asus\Desktop\mlops-recommender> python src/test_data.py
4       1      356     5.0  1225735119

Shape of dataset: (100000, 4)
(venv) PS C:\Users\Asus\Desktop\mlops-recommender> python src/data_transform.py
Data Transformation Completed!

   user_id  product_id  rating   timestamp interaction
0        1           1     4.0  1225734739    purchase
1        1         110     4.0  1225865086    purchase
2        1         158     4.0  1225733503    purchase
3        1         260     4.5  1225735204    purchase
4        1         356     5.0  1225735119    purchase
(venv) PS C:\Users\Asus\Desktop\mlops-recommender>
```

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OUTLINE

TIMELINE

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src > test_data.py > ...

```python
import pandas as pd

# Load dataset
ratings = pd.read_csv("data/ratings.csv")

# Take subset for performance
ratings = ratings.head(100000)

print("Dataset Loaded Successfully!\n")
print(ratings.head())

print("\nShape of dataset:", ratings.shape)
```

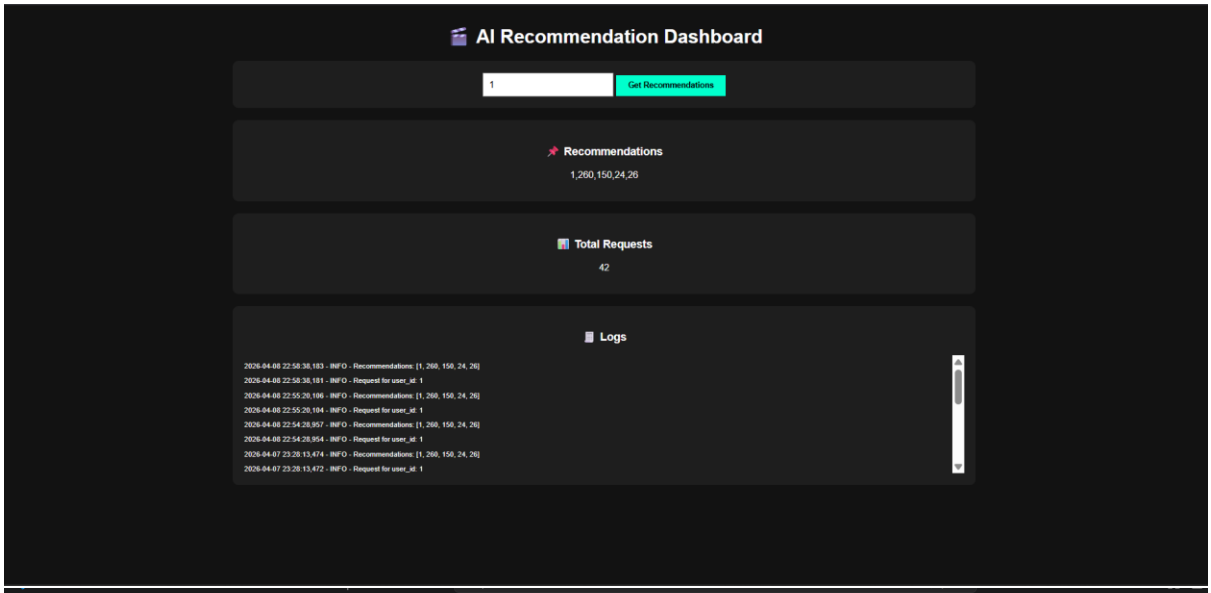
PROBLEMS | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS

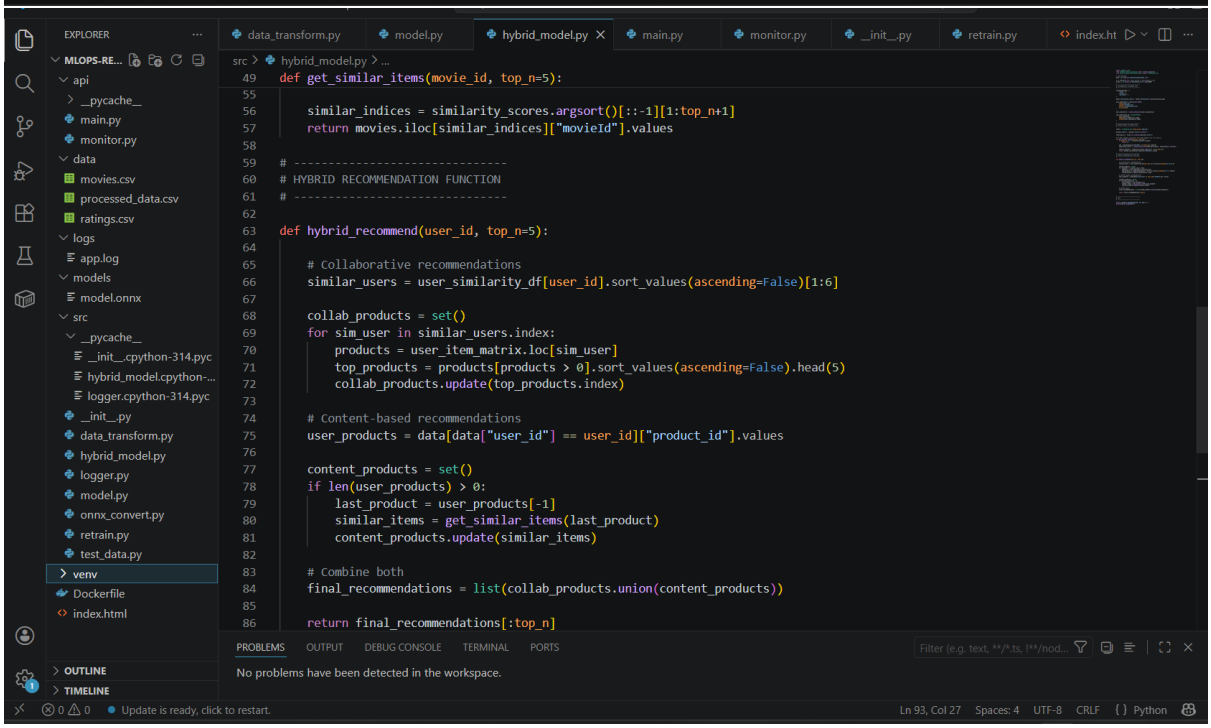
History restored

```
PS C:\Users\Asus\Desktop\mlops-recommender> & c:\Users\Asus\Desktop\mlops-recommender\venv\Scripts\Activate.ps1
(venv) PS C:\Users\Asus\Desktop\mlops-recommender> python src/test_data.py
Dataset Loaded Successfully!

   userId  movieId  rating   timestamp
0       1        1     4.0  1225734739
1       1      110     4.0  1225865086
2       1      158     4.0  1225733503
3       1      260     4.5  1225735204
4       1      356     5.0  1225735119

Shape of dataset: (100000, 4)
(venv) PS C:\Users\Asus\Desktop\mlops-recommender>
```


AI Recommendation Dashboard
1
Get Recommendations
Recommendations
1,260,150,24,26
Total Requests
42
Logs
2026-04-08 22:58:38,183 - INFO - Recommendations: [1, 260, 150, 24, 26]
2026-04-08 22:58:38,181 - INFO - Request for user_id: 1
2026-04-08 22:55:20,196 - INFO - Recommendations: [1, 260, 150, 24, 26]
2026-04-08 22:55:20,194 - INFO - Request for user_id: 1
2026-04-08 22:54:28,957 - INFO - Recommendations: [1, 260, 150, 24, 26]
2026-04-08 22:54:28,954 - INFO - Request for user_id: 1
2026-04-07 23:28:13,474 - INFO - Recommendations: [1, 260, 150, 24, 26]
2026-04-07 23:28:13,472 - INFO - Request for user_id: 1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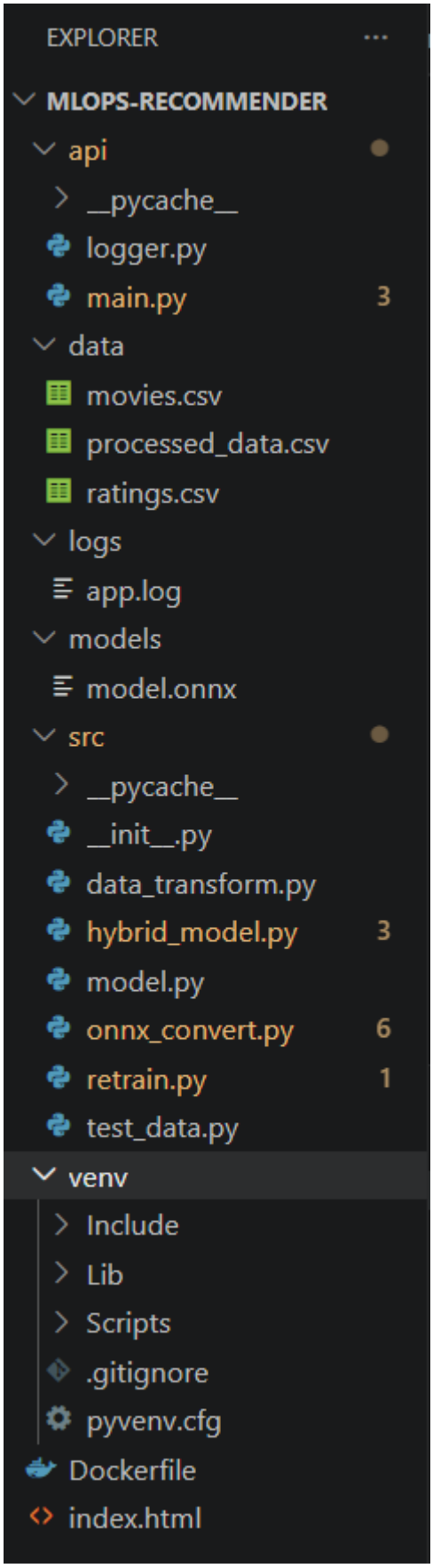
index.ht
src > hybrid_model.py > ...
49 def get_similar_items(movie_id, top_n=5):
55
56 similar_indices = similarity_scores.argsort()[::-1][1:top_n+1]
57 return movies.iloc[similar_indices]["movieId"].values
58
59 # ------------------------------
60 # HYBRID RECOMMENDATION FUNCTION
61 # ------------------------------
62
63 def hybrid_recommend(user_id, top_n=5):
64
65 # Collaborative recommendations
66 similar_users = user_similarity_df[user_id].sort_values(ascending=False)[1:6]
67
68 collab_products = set()
69 for sim_user in similar_users.index:
70 products = user_item_matrix.loc[sim_user]
71 top_products = products[products > 0].sort_values(ascending=False).head(5)
72 collab_products.update(top_products.index)
73
74 # Content-based recommendations
75 user_products = data[data["user_id"] == user_id]["product_id"].values
76
77 content_products = set()
78 if len(user_products) > 0:
79 last_product = user_products[-1]
80 similar_items = get_similar_items(last_product)
81 content_products.update(similar_items)
82
83 # Combine both
84 final_recommendations = list(collab_products.union(content_products))
85
86 return final_recommendations[:top_n]
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
No problems have been detected in the workspace.
Update is ready, click to restart.
Ln 93, Col 27
Spaces: 4
UTF-8
CRLF
Python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MLOPS-RECOMMENDER
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